ABSTRACT:

Though progress has been done identif faces, today's tech struggles when faced with massive photo sets. Earlier work brought forward FaceNet a method shifting facial information into structured space where likeness closely follows measurable gaps. Within this new structure, distinguishing individuals, verifying who they are, or clustering portraits becomes more straightforward using simple methods on FaceNet's generated values. This shift altered how visual comparisons rely on numerical distances.

Putting together the deep convolutional setup involved blending varied versions of the main design into a single framework. Rather than using a hidden layer, training guided the model to connect predictions directly to correct outputs. The shift lay in signal flow - no intermediaries, simply mapping input straight to result.

A different kind of preparation kicked things off - using three versions of cut-out faces, where similarity wasn't always guaranteed. Choices in cropping followed what actually performed well during real-time runs. The system moved faster since the core identification technique uses only 128 bytes for each face, a detail mentioned before. Something new appears now: results almost always land at 99.63 percent pushing this method near the top of the LFW rankings.

CHAPTER 1

INTRODUCTION

1.1 PROJECT DESCRIPTION

One way to handle roll call starts with spotting faces instead of names on paper. A camera snaps photos without delay, catching whoever steps into view. From there, software checks each photo, isolating any visible face almost instantly. That captured face then lines up against a collection of saved ones, looking for a close match. When it finds someone familiar, the system logs who showed up, along with when they appeared. Names get added quietly, no typing required, just quiet precision behind the scenes. Mistakes fade away because machines track presence more carefully than humans flipping pages. Less effort flows from staff, while accuracy grows in silence. Records stay clean, built moment by moment.

Out here, computer vision isn’t just theory - it shows up in real tasks. Instead of separate steps, image handling flows into clever code, building a way to track presence without old flaws. Most current systems rely on paper or simple digital logs, yet they slip up easily - mistakes happen, people mark names for others, time drags. With this method, precision tightens while effort drops. Face scanning now does more than identify; it reshapes how roll call works. Old habits fade when tech skips delays and cuts false entries. Accuracy sticks around because the design learns patterns, not just records them. Efficiency climbs since checks speed up without extra staff. What used to take minutes folds into seconds. Reliability grows because the setup adapts, unlike rigid earlier versions.

One way to spot faces in photos? A system built here leans on Convolutional Neural Networks - deep learning tools good at seeing what's inside pictures. These networks grab details like curves, lines, and textures straight from images, no human setup needed. Because they learn on their own, mistakes happen less often than older methods had. Even if light shifts, someone frowns, or a head tilts oddly, it still works well.

A camera takes pictures of faces, then software studies them closely. After seeing many examples, the neural net begins recognizing people through subtle details. When someone appears live, it matches the face quickly, attaching correct timestamped records. No need for staff; decisions happen without interference. Mistakes fade because there is no room for tricks or gaps.

0When CNN-based face detection meets log recording, automation gains safety, zero contact, one swift step - perfect for today's classrooms or offices. This effort shows deep learning can solve actual problems while pointing toward smarter tools that save moments, reduce errors, fewer tasks piling up behind the scenes.

1.2 OBJECTIVES

• This project's ultimate aims put an intelligent face recognition attendance system in place. The attendance automatically marks itself with convolutional neural networks.

• A modern, accurate, contactless solution for the existing traditional manual and semi-automatic attendance is needed to improve efficiency and reliability.

• The latest cutting-edge objective is to create a CNN-based facial recognition model that can accurately identify distinctive features on the face.

• This system is designed to cope with typical real-world difficulties, including variations in facial expressions, poor lighting conditions and camera angles. Its performance under such typical difficulties remains constant though.

• It is also another important goal to get rid of proxy attendance and unauthorized entry so that people are forced to identify themselves by biometric measures whenever they participate.

• Distinguish between card identification or traditional attendance systems and use facial recognition identifications which reduces authenticity.

• The project aims to make the entire attendance process fully automatic, including face detection, recognition and attendance record creation.

• All attendance details, such as the time, name and date, are stored automatically. This makes for less work in theory and fewer mistakes when entering data

• The goal is to make the system easy operating for administrators who have little technical knowledge.

• Scalability is one of the major goals, designed to adapt the system to higher or differing levels and it can be used in many different organizations including educational institutions as well as enterprises.

• The project applies deep learning and computer vision in real world administration mechanisms.

• It gives a solid base for future improvements to include such things as cloud integration, mobile access and improved security.

• Efficient real-time performance is a further important aim, with attendance registration taking place without inappropriate slipping.

• System no disruption to normal work during operation.

• The focus of the system is to ensure data accuracy and consistency in the long run.

1.3 PROBLEM STATEMENT

Right away, this tech works on anyone out in public without slowing things down. Still, worries about privacy or unfair treatment drop because pictures get used just to spot someone - nothing else, no storage. Even with massive piles of data, smart methods and modern math boost how often matches are correct.

Besides those points, entrance surveillance cameras might serve this function too. That cuts setup expenses while boosting building safety - existing devices handle the job. Privacy questions fade because the hardware stays regardless; face scans apply strictly to tracking presence. Overall, people find it simpler compared to alternatives - it leans less on manual effort. Yet reliability climbs here, thanks to quicker processing and fewer slipups caused by individuals.

Put simply, this kind of setup helps smooth things out while producing reliable records on who showed up. Yet here's the catch - even though automatic check-ins mark progress, putting them in place won't wipe away issues tied to real presence at lectures or jobs. What really matters is this: showing up isn't just about gadgets, it lives inside how people choose to act. Schools and colleges now using these tools find fewer worries around accuracy since experience has already cleared many early doubts.

Still, making a working facial recognition setup for tracking attendance isn’t without its problems. Lighting shifts, how someone is feeling, head angles, partial coverings, or blurry images can all weaken accuracy. Old-style machine learning methods struggle here - since they depend on handpicked features to detect changes. Because of this, systems that learn complex face details on their own are now needed. A smarter approach must step in where older ones fall short.

One way machines learn to spot faces? Through Convolutional Neural Networks - deep learning tools built mainly for image tasks. These networks pull useful details straight from pictures without needing manual help, often nailing tough scenes where lighting or angles aren’t ideal. Still tricky though: getting the setup right means watching how models train, what data they see, whether it runs fast enough live, and if the whole thing grows smoothly with more users.

So here's the core challenge: building an attendance setup that runs without fail, works on its own, uses CNN tech, and skips past what older systems can’t handle. It needs to spot individuals right away while shutting down fake check-ins, cutting manual labor, plus keeping logs safe and neatly sorted. On top of that, it ought to adapt easily, scale up when needed, stay simple to operate, and slide into spots like classrooms or workplaces without hassle.A fresh approach steps in where old methods fail, bringing smarter ways to track who’s present without missing what schools actually need. This setup sidesteps earlier problems while matching how tech moves today.

CHAPTER 2

LITERATURE SURVEY

2.1 EXISTING AND PROPOSED SYSTEM

In the past years there has been a noticeable rise in the requirement for automated and smart attendance systems, which is mainly because the drawbacks associated with the manual methods like attendance registers, punch cards, and ID-based systems. These traditional practices consume a lot of time, are susceptible to human errors and guarantee the practice of proxy attendance. This, in turn, led to the researchers and developers looking into biometric techniques as a means to improve the attendance management systems in terms of both accuracy and efficiency. Among the different biometric approaches, face recognition has become the most favored one since it is non-invasive, user-friendly, and does not require any physical interaction with the subject.

Moreover, along with the development in image processing and computer vision, face detection and recognition techniques have matured considerably. The first generation of face recognition techniques made use of very simple geometric characteristics, for instance, the distance between the eyes, nose shape, and jaw structure. Even though these techniques were easy on the computation, they were very dependent on changes in lighting, facial expressions, and head pose. Therefore, their performance was very limited in real-life situations, such as classrooms and offices.

Deep learning is now considered to be a pivotal factor in the development of face recognition systems. Neural Networks with Convolutional layers (CNNs) have been the ones that got the most influence as they can detect and identify human faces in the most complex manner through learning from images. CNN-based models also come with the benefit of large-volume datasets handling and high accuracy rates even under the most difficult lighting, facial expressions, and occlusion scenarios. It has been pointed out that deep-learning-based systems for face recognition are far more robust and scalable

2.1.1 PROPOSED SYSTEM

Right away, facial recognition steps into the spotlight for tracking who shows up. This setup skips old methods like signing sheets, swiping cards, or scanning fingers. Instead of relying on those, it leans on unique facial features to mark presence. Efficiency climbs when faces become the key. Accuracy tightens because each person must be physically there. Faking attendance fades since someone else cannot stand in. Built without a backdoor for substitutes, it holds firm. No shared logins, no borrowed identity tricks. Every check-in ties directly to a real-time face. Smooth operation follows from cutting manual entries. Mistakes drop when machines handle matching. What emerges is simpler, sharper, more reliable. Presence becomes something you show - not something claimed.

From a video feed, the camera captures how people look on their faces - say, learners at school or workers at an office. After that, software steps in to work through those pictures using digital methods. First thing it does: search for any human face inside the frame. That moment when the system spots a person's face marks the start of reading emotions from what they show visually. Out of the frame comes just the face, pulled tight around the expression. This cut isn't random - it sharpens what matters in the picture. Resizing follows, adjusting pixels to fit a standard shape. After that, pixel values shift into a common range through normalization. Each step quietly lifts how well expressions are later recognized. Precision grows behind these small adjustments. Without them, reading faces would be far less reliable.With eyes on accuracy, facial details like eye placement or nose shape get spotted by a network trained to notice them. A collection of member photos forms the foundation, feeding data so patterns can emerge over time. Instead of guessing, the system learns directly from examples stored in its structure. Recognition happens when similarities align between new input and known traits pulled from past exposure.

When enough likeness shows up - clear matches only - their presence gets logged automatically. Time stamps lock in once identity confirmation feels solid. Matching strength decides whether entry goes through cleanly.

A locked database keeps user information safe. Because only staff members and teachers can change or see records, mistakes stay low. This setup means details remain correct without unwanted changes sneaking in. No paperwork happens here - everything lives online instead. Fewer people need to step in while things run on their own.

2.2 FEASIBLITY STUDY

What happens next for the proposed Face Recognition Attendance System depends on a close look at how well it works alongside tight resource limits. Examining technical fit comes first - then costs slide into view. Operations matter just as much, with day-to-day function under scrutiny. Laws and rules shape part of the check too. Every piece builds a picture of whether the system holds up when tested beyond theory.

2.2.1 TECHNICAL FEASIBLITY

The system had been proposed and is technically feasible because it is basis on the use of some techn that are well known and widely utilized such as digital cameras, image processing techniques, and deep learning algorithms. Modern-day computers and laptops are well fitted for the good performance of face detection and recognise tasks. There are various open-source libraries and frameworks that help in developing Convolutional Neural Network models, thus making the implementation easy even for educational institutions. The fact that the system can be run on a standard computer without the need for specialized devices, managing technical complexity becomes easy.

2.2.2 ECONOMIC FEASIBLITY

Looking at it from an economic angle, the system is really cheap. One thing leads to another when paper logs disappear - no more pens, no more stamps. Instead, machines track who shows up, using fingerprints or radio tags, though those add price tags fast. Cameras take center stage now; they cost something, yes, but not much compared to old ways. A desktop machine does the thinking, running programs built without paying licensing fees. Work piles shrink because fewer people must sort through sign-in sheets each day. Over months, hours saved turn into money saved too. Schools find this useful, especially ones watching every dollar. Smaller groups benefit just as clearly, even if they start slow.

2.3 LITERATURE REVIEW

Adapted from ANIL K JAIN,

Echoing his article, face recognition-based attendance systems have attracted people's attention as a set of new-tech, contact-free solutions that replace old manual ways for marking time or taking roll by computer voice notes. These systems use the latest in computer vision and deep learning to identify people instantly, and to clock attendance with an unprecedented speed and accuracy. Research contributions done by Anil K. Jain have had an important impact on the design of biometric recognition systems, particularly by stressing the crucial importance of featuring representation and matching means. Work he has done also deals with common challenges that face the biometric recognition of faces, for example changes in lighting conditions, facial expressions, pose variation failures and differences within one individual over time.

Inspired by these foundational ideas, some recent publications apply convolutional neural networks to learning directly from images discriminative facial features. Many of these deep learning models use some form of classification or similarity-based matching in order to improve performance across a variety of real-world environments. In addition to the complex convolutional nets, some researchers emphasize the need for preprocessing: face detection, alignment and image normalization. Such steps can suppress noise and make for an accurate model.Prevention of fraudulent attendance from spoofing attacks is another important research topic. Liveness detection techniques, texture analysis and motion-based verification are all popular ideas for improving system reliability.

Nevertheless, the current literature strongly suggests that if effective preprocessing, feature learning based on CNNs, and security mechanisms are combined then reliable, accurate and scalable face recognition attendance systems can indeed be built - and are suitable for application in the real world.

2.4 TOOLS AND TECHNOLOGIES USE

A single mismatch won't stop progress, still every component counts in a facial attendance system. Starting at square one, decisions guide image handling, face detection, data storage, user interaction. A weak link could drag performance - thoughtful picks keep flow steady. Gadgets work out of sight, running quiet, asking nothing.

2.4.1 PROGRAMMING LANGVAGE

The choice of the main programming language for the project's implementation is Python. Its clear syntax, readability, and its solid backing for machine learning and image processing make it the most popular language choice. Python also supports faster development and module integration with less effort in the case of face detection, recognition, and database management..

2.4.2 MACHINE LEARNING AND DEEP LEARNING

For the implementation of face recognise deep learning methods are used, mainly Convolutional Neural Networks (CNN).

Image Process Libraries

Starting off, handling pictures plays a key role when spotting and naming faces. Computer vision tools help manage this task by pulling visuals straight from the camera feed, outlining facial areas, then running adjustments like scaling down, shifting to grayscale, plus tweaking light levels. Each step sharpens the picture clarity while giving the identification system a boost in accuracy. Finished.

DATABASE TECHNOLOGY

Storing user info, facial encodings, and check-in logs happens inside a database setup. For school-related builds, lean databases work smoothly - retrieval speed meets simple upkeep. Timestamped entry logs stay safe, structured, neatly arranged thanks to how the system holds data.

HARDWARE COMPONENTS

A digital camera or webcam captures face images live. Running on regular computers, the software works best with solid performance specs. Recognition happens right there during operation. No expensive gear needed makes setup low cost. Simple installation keeps everything accessible without hassle.

DEVLOPMENT ANALYSIS

A workspace built for coding helps craft, check, run, test, fix issues - all within one place. Features such as smart tips while typing, spotting mistakes early, tracking file changes boost how fast work moves forward.

OPERATING SYSTEM

While setup might differ slightly, adaptation stays straightforward regardless of the base system in play.Fundamentally built on thoughtful choices, the tech behind this attendance system handles face recognition smoothly across schools or workplaces. Its design grows with demand while staying practical in daily operation. Performance stays strong under varied conditions thanks to well-matched components. Real environments test it best - here, accuracy meets reliability without slowing down.

2.4 SYSTEM REQUIREMENTS

Hardware Requirements

- **Processor:**
  Intel Core i3 or equivalent (minimum)
- **RAM:**
  4 GB RAM (minimum)
- **Hard** 100 GB free storage space
- **Camera:**
  HD Webcam (720p or higher recommended)
- **Input:**
  Keyboard and Mouse
- **Output** Monitor or Display Screen

Software Requirements

- **Operating**
  Windows 10 / Windows 11 / Linux (Ubuntu or equivalent)
- **Programming:**
  Python 3.8 or above
- **Development**
  Visual Studio Code / PyCharm / Jupyter Notebook
- **Libraries & Frameworks:**
  - OpenCV (for image processing and face detection)
  - NumPy (for numerical operations)
  - Deep Learning Framework (for CNN implementation)
  - Face Recognition Library (for feature extraction and matching)
- **Database:**
  SQLite / MySQL (for storing user and attendance data)

CHAPTER 3

3.1 SOFTWARE REQUIREMENT SPECIFICATION

3.2 Purpose

The purpose of this Software Requirement Specification (SRS) document is to describe the functional and non-functional requirements of the Face Recognise Attendance System. This document serves as a reference for developers, testers, and users to understand the system's behavior, features, and constraints.

3.3 SCOPE

The Face Recognise Attendance System is designed to automate the process of attendance marking using facial recognition technology. The system captures facial images through a camera, identifies individuals using deep learning techniques, and records attendance automatically in a database. The system aims to reduce manual effort, prevent proxy attendance, and improve accuracy and efficiency.

3.4 Functional REQUIREMENTS

- The platform shall capture facial images using a camera.
- The model shall detect faces from the captured images.
- The structure shall recognize registered faces using a trained CNN model.
- The system shall mark attendance automatically upon successful recognition.
- The model shall store attendance records with date and time.
- The platform shall authorized users to view reports.
- The platform should recognize faces within a few seconds.
- The structure should support multiple users in a single frame.

3.5 Non-Functional Requirements

3.5.2 Security Requirements

- Only auth users should access the system.
- Facial data should be stored securely to prevent misuse.

3.5.3 Usability Requirements

- The system should have a simple and user-friendly interface.
- Minimal training should be required to operate the system.

3.5.4 Reliability Requirements

- The system should function accurately under normal lighting conditions.
- The system should maintain data consistency and integrity.

3.6 Method

This pertains to the explanation of the processes utilised during the extraction of the face images of individuals and the notation of the attendance of the individuals within the system.

The process involves image capture. In this context, the images of the registered staff in the organization were captured using a digital camera or a webcam. Different images of the staff were taken, considering expression and lighting variations. This was done to ensure the efficacy of the facial recognition system. The captured image was saved in a system database with the intent of training the facial recognition system. The next procedure involves image processing. The primary aim of this step was to enhance the image quality. This was made by resorting to the resizing of th img to a standard size prior to the image transformation process. Converting the image to greyscale is followed if necessary.

The next method shall be face detection. This can be considered as the step in the process where the system also has the capability to determine the presence and/or position of the faceof the human subject within the image. This canbe considered as the step in the process where the only part of the img that has the diminish the effect of the image that can come from the fact that it has the face from those pictures, work moves on to spotting key details through a system built for pulling out and identifying traits. To handle this job, a method called Convolutional Neural Network - CNN for short - comes into play. What it does is take in facial visuals, then map distinct elements like eye shapes, overall face structure, along with outline patterns. Instead of just seeing pixels, the model learns what makes each face stand apart by focusing on these specific markers. During training, the system begins learning unique traits from the stored face picture. When testing runs, those detected traits act like a coded version of the person's face.

Right after facial recognition finishes, the system marks attendance along with how sure it needs to be about the match. Triggered without any manual step, the database handles this part on its own. Using who showed up, plus the exact day and hour, it logs everything quietly. Because of this setup, one person can’t check in twice for the same session. Each entry sticks only once, locked by time and identity both.

One last thing - the setup includes a way to track who shows up, visible only to approved users through the interface. That means signing in without needing to touch anything.

Looking at how well the system performs helps fine tune how sure it is about what it recognizes. When certainty goes beyond a set limit, that data should stay out. Keeping wrong details from entering matters most in this area. Proper light conditions must be met for clear images during capture. Camera placement adjusts itself so lighting works best. The updating processes of the information in the facial database system would take place in the context of the changes that would result because of the impacts of time. The system would also take into consideration the measures that would be put in place to reduce the posed problem due to the failed camera and database system.

CHAPTER 4
SYSTEM DESIGN

The Face Recognition Attendance System's design description covers the solution is built, how its parts interact, and how data made from one end to the other. Its design centers around four main aspects: automation, accuracy, security, and usability, along with the option of future improvements. Each part of the model is designed to work separatly from one another, following a modular approach as the whole system works in synchronization.

1. Design Overview The Face Recognise Attendance System is designed to be an independent application that integrates several functionalities, such as capturing images, analyzing faces, recognition logic, and data storage. The system goes through a number of stages starting from image capture to the final attendance record generation. Each stage is logically separated from the other so that complexity is reduced and maintenance is easier. The system is designed to function in real-time environments such as classrooms or offices. The model is designed to slow the human involvement in marking the attendance which in turn will reduce the errors and manipulations.

2. Image Acquisition Module The image acquisition module takes responsibility for capturing the live video frames through a webcam or an external camera. Right when timing begins, live video feeds pour into memory so every person there shows up sharp and full in view. One snapshot can hold several faces at once, no need to wait between them. Before anything runs, hardware tests run first - specifically looking for active camera links. When no device answers or acts broken, warnings pop up immediately, stopping next steps by design. That early gate keeps records trustworthy, blocking false entries before they start.

Usually, photos snapped by a camera need fixing before anything useful can be done. These pictures often carry noise or uneven lighting that muddles details. Changes in surroundings - like what's up front or behind - affect clarity too. Camera-specific quirks such as resolution shifts or setting mismatches add more hiccups. Without adjusting for these, results go off track.

4.2 SYSTEM PERSPECTIVE

The system perspective not only illustrates the Face Recognition Attendance System's integration into the whole operational environment but also clarifies its engagement with different kinds of users, hardware, software, and data. This part of the document provides an abstract presentation of the system, where the interaction among various components is the spotlight rather than the description of the internal algorithms. The proposed model is a complete app that leverages facial recognition technology to automate attendance management.

Looking at it widely, the system substitutes the conventional attendance techniques like the physical registers and card-based systems. It is a single-point of processing that recognizes the face data, does the smart filtering, and records the attendance without any human involvement. The system is for standalone operation, but later on, it could be linked or combined with other academic or organizational management systems for the future.

4.2.1 System Environment

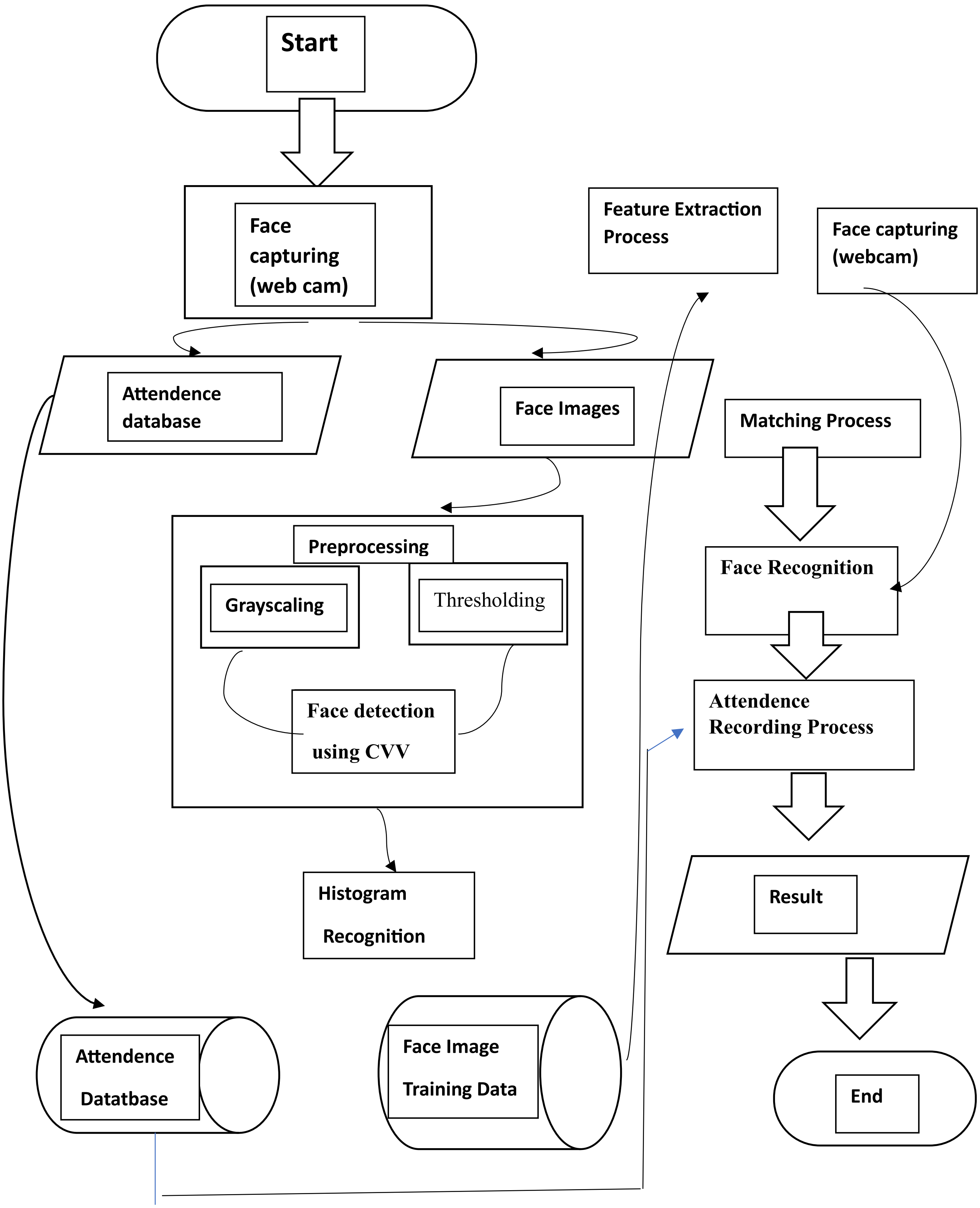
The Face recognise Attendance System has the ability to function under such a controlled indoor environment as classrooms, labs, offices, or training centers. It is a camera-enabled computing device that the system needs to operate. The system is intended to be functional in normal lighting conditions and in a standard classroom setup. One of advantages it has over its competitors is it do not rely on specialized biometric hardware, making it a solution that can easily be integrated into the current infrastructure.

Moreover, the model is capable of running on popular operating systems, Windows or Linux, and connecting to standard peripherals like webcams, keyboards, mice, and a display unit. This enables the institutions to implement the model with no or little additional cost and effort terms of hardware changes.

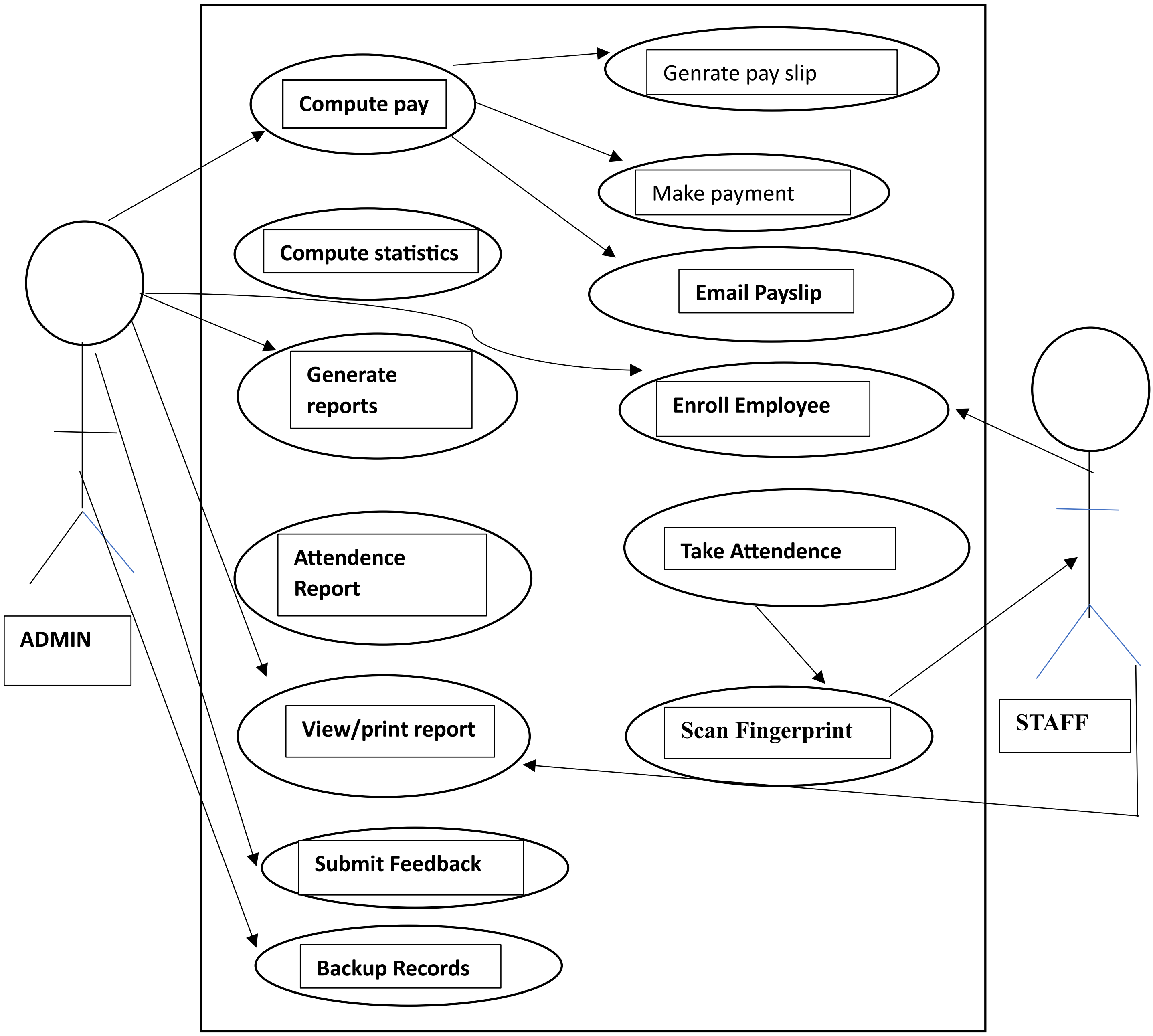
CHAPTER 5

DETAILED DESIGN

5.1 ER DIAGRAM

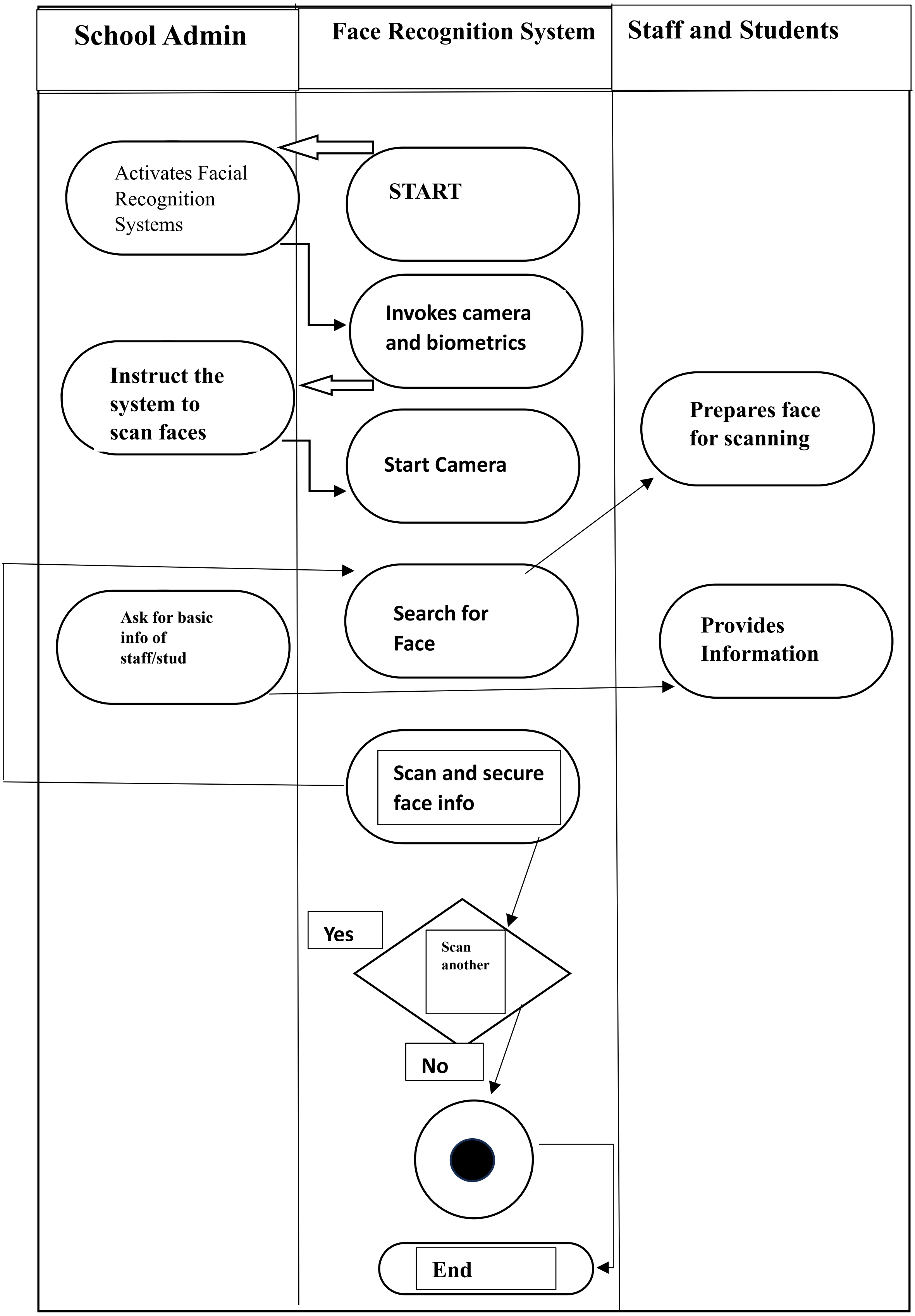


5.2 USE CASE DIAGRAM FOR ATTENDENCE SYSTEM

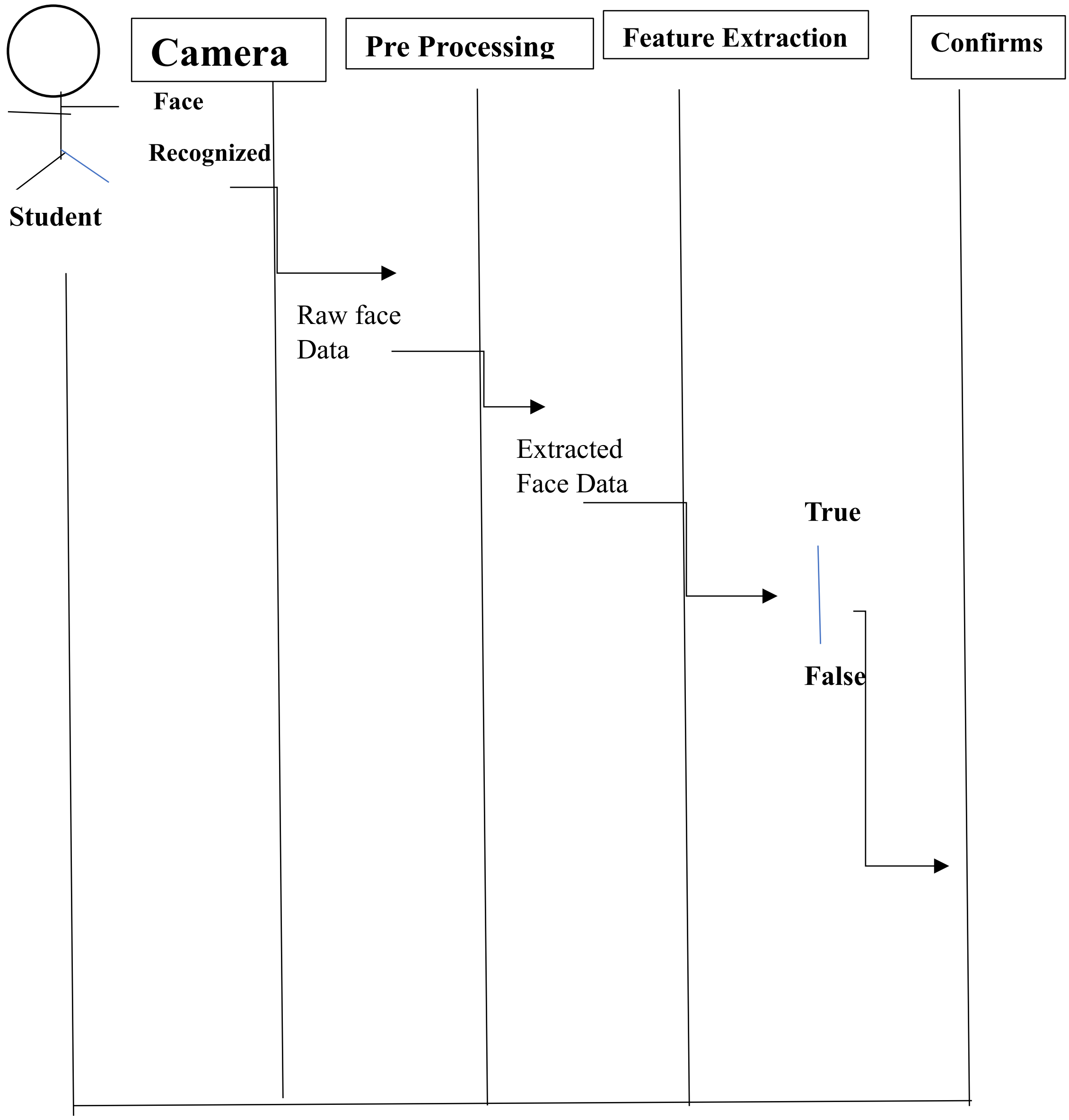


- Use casee diagram for Face Recognition Attendence System
- The ER diagram represents entities such as Student, Face Data, and Attendance, along with their attributes and relationships.
- Each student is linked to unique facial data, which is used to identify and verify the student during attendance marking.
- The attendance entity stores details like date and time and is associated with the corresponding student for accurate record keeping.

5.3 ACTIVITY DIAGRAM FOR FACE RECOGNITION ATTENDENCE SYSTEM



5.4 SEQUENCE DIAGRAM



- At the beginning, the student stands in front of the camera, allowing the system to capture their facial image. This image is then processed to remove unwanted noise and improve clarity so it can be analyzed accurately.
- Once the image is prepared, the system identifies important facial characteristics needed for recognition.
- These things are checked against the records already stored in the database. When a valid same is found, the system verifies the student's identity and records the attendance without any manual involvement.

CHAPTER 6

IMPLEMENTATIONS

The Face Recognition Attendance System is introduced through a well-planned and modular manner that combines computer display and deep learning techniques. The system is made to detect people by their facial features and at the same time very accurately register their attendance throughout the day automatically. The whole process of implementation is breakup into main stages: data collection, preprocessing, CNN-based model development, face recognition and attendance recording.

1. Data Collection

The implementation starts with the gathering of the personal image data. Different lights, faces and angles are changing to create conditions for capturing the subjects and to fortify the model. By taking multiple photos of each person, CNN is better prepared to distinguish individual facial characteristics. The images of the person are given a unique ID or name and the images are placed in folders labeled according to the training purpose.

2. Image Preprocessing

Subsequently, the pre-processed images are working text for the CNN model's training. The procedure boosts the recognition accuracy using OpenCV. The computational load is reduced by converting the images to grayscale which helps in keeping good facial features. The background is removed from the facial region through a face detection operation done with pre-trained classifiers. The faces that are detected are resized to a common size so that consistency across the dataset is maintained. More preprocessing steps are performed like normalizing and reducing noise as they improve the quality of input data and guarantee uniformity during model training.

3. CNN Model Development

- The implementation of the Convolutional NeuralNetwork forms the basis of the system. The CNN model is created automatically and effortlessly fetch the significant facial features present in the input images. Generally, the architecture has several layers, which are followed by activation functions and pooling layers. The convolutional layers perform the task of recognizing both low-level and high-level facial features – edges, contours, and textures. The pooling layers minimize the spatial dimensions and, at the right time, make the computations more efficient.

- At the last of the CNN, there are the fully connected that conduct the classification basedon the features that has been learned. The model are trained with the labelled facial dataset where the CNN learns to link the feathures of faces with their respective identities. During the teaching phase, the dataset gets split into the teaching and validation sets, in orderto assess the system performance and prevent overfitting. The training period goes on until the desired accuracy is reached.

4. Face Recognition Process

After the CNN system has been trained, it gets the chance to be a part of the real-time recognition module. The camera takes live video frame that are processed one by one. In each frame, faces are recognized and their preprocessing follows the same procedure as the training images. Then the processed face image is sent to the trained CNN model, where it predicts the person's identity.

- If the predicted identity is that of an enrolled individual and the confidence level is high enough, the system acknowledges the recognition. If no identity is found, the system either disregards the face or labels it as unknown. This real-time recognition procedure guarantees that only the registered and authorized individuals are considered present.

5. Attendance Recording Mechanism

The process of attendance marking is initiated after the recognition has been accomplished successfully. The system automatically records the person’s name, date, and time. A structured format like a CSV file or a database is used to store the attendance data which makes it retrievable and analyzable later on. The system checks if the person’s attendance is already marked for the current session before updating the record in order to eliminate duplicate entries.

- The whole process is automated which means no manual attendance marking is done and hence, errors are greatly reduced. It also guarantees that attendance records are transparent and accurate.

6. System Integration and User Interaction

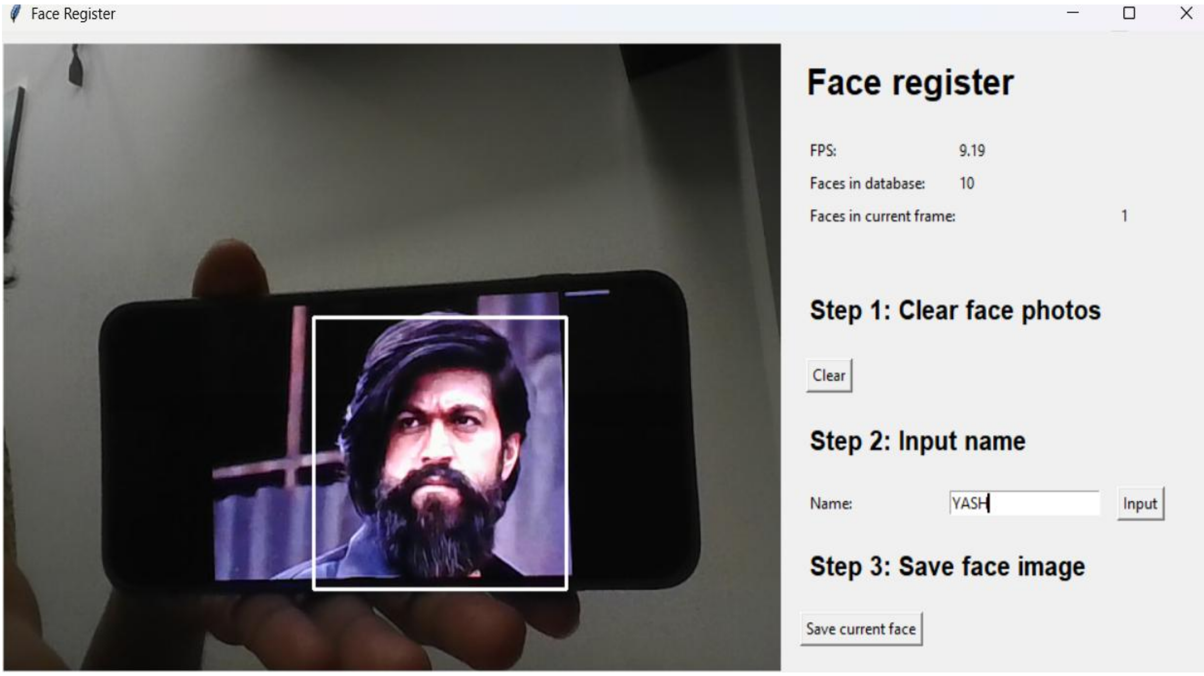
The entire system combines camera input, CNN-based recognition, and net storage into a single operation. A straight forward interface is offered for user enrollment management, recognition status monitoring and attendance logs viewing. The model is designed to been user-friendly so that the administrators can run it without needing to have comprehensive technical expertise.

7. Performance and Reliability Considerations

The implementation is centered around achieving real-time performance with very small delay. Optimization methods like effective preprocessing, model tuning, and hardware using are employed to keep things running smoothly. Moreover, the system is put through various testing conditions to ensure its reliability and accuracy.

CHAPTER 7

SCREENSHOTS

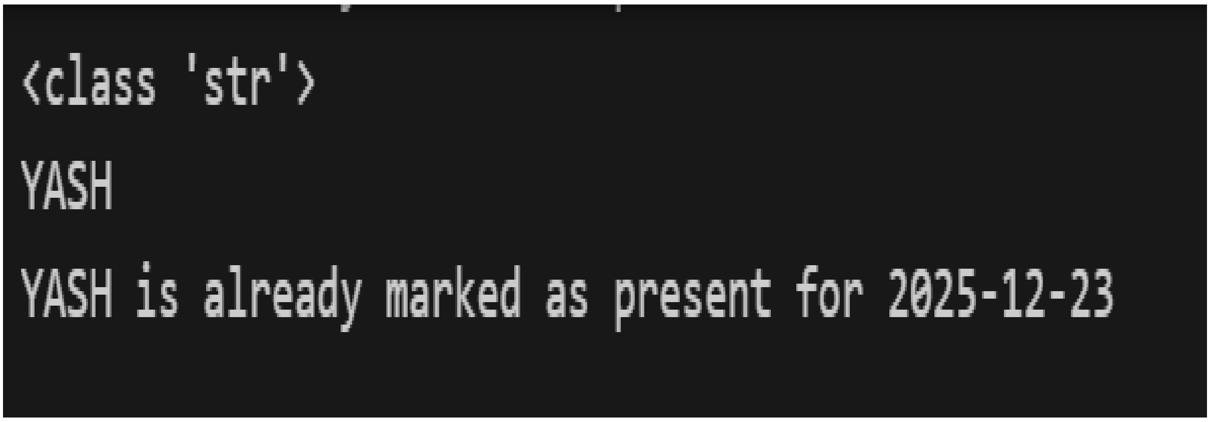


Face Input is given.

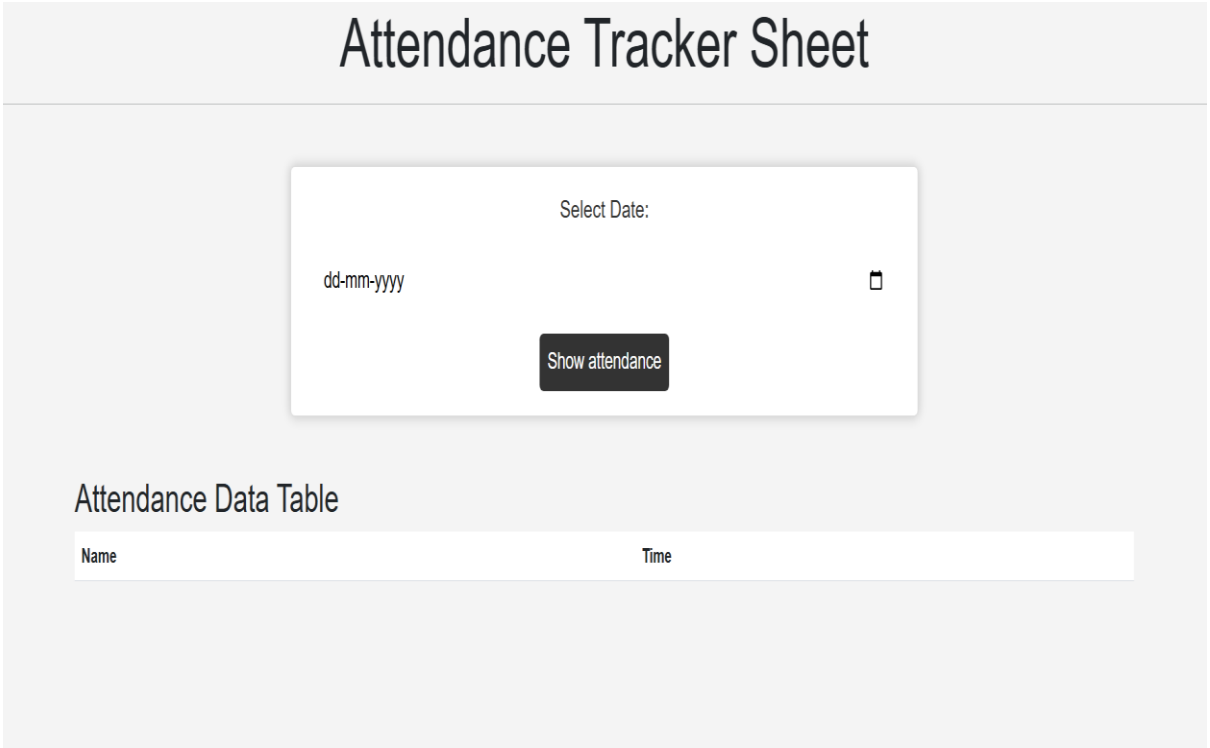
Standing before the camera, a person offers their face as the input. Their presence is captured through stillness in that moment

Face Recognised.

Facingt camera, the person waits for their features to be spotted. A still position helps the system notice facial points. Lighting matters when the software begins its scan. Recognition kicks in once key markers are found. Movement can blur the outline the tech needs.



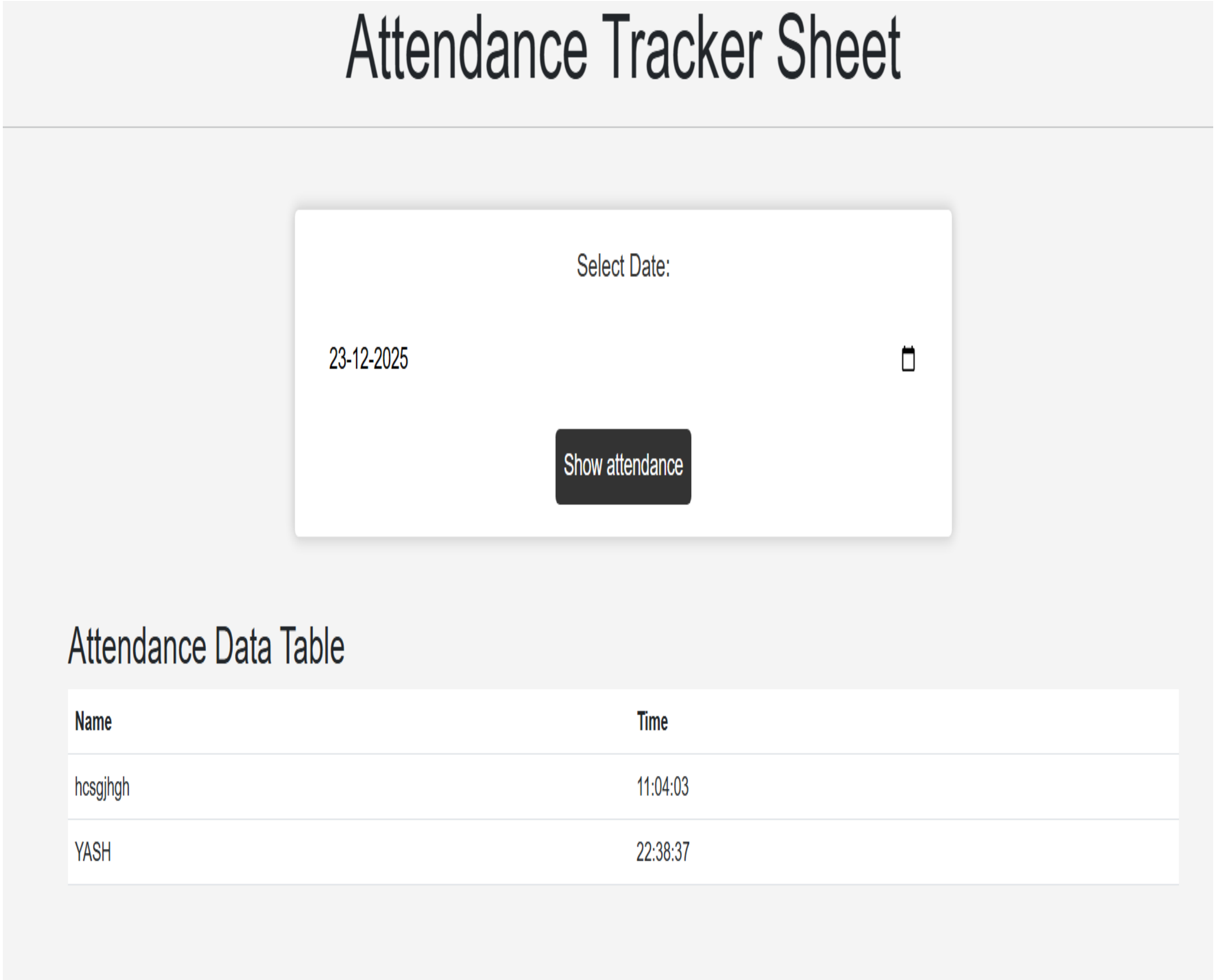
Recognised and marked as present.



Attendence Tracker Dashbourd.

• The attendance tracker dashboard provides a centralized view of all attendance records in an easy-to-understand format.

• Users can be able to select specific date from the calendar to view attendance details.for that particular day.



Marked as present

• Once a date is selected, the dashboard displays the list of students who were present on that date.

• Each record includes important details such as student name, identification number, and time of attendance.

CHAPTER 8
SYSTEM TESTING

8.1 SYSTEM TESTING

System testing represents a major milestone in the route to the completion of the Face Recognition Attendance System and at the same time it is the assessment of the whole and integrated application to guarantee its conformity with the stated requirementsRight from the start, each part of the project - face detection, using CNN for recognizing faces, recording who's present, and saving information - is tested together as one whole piece. What drives this phase is making sure everything runs smoothly, delivering outcomes that stay correct and steady every time.

First off, check if the camera works by confirming it sends live video smoothly, no lag or missing frames. After that comes face detection - test how well it spots faces when lighting is bad, surroundings are messy, or people tilt their heads. Then see if multiple faces show up at once, each marked correctly in their own space.

Afterward, a face detection tool using CNN technology checks both known and unknown individuals.This ensures consistent performance under real conditions. Finally, the check-in process gets reviewed to verify nobody gets logged more than one time during a single meeting. After checking how the system behaves, attention turns to whether names, dates, and times are saved properly - either into a database or a CSV file. When repeated records appear, they get flagged before steps block repeats, keeping information uniform. Consistency matters most once storage paths are confirmed working as intended.

From start to finish, the whole setup gets checked for speed and how well it runs. Performance under pressure reveals whether face detection keeps up instantly. When things go sideways - like blurry video or wrong data - the response shows if stability holds. Glitches enter the scene on purpose, just to see recovery happens smoothly. No hiccups should show when lighting fails or someone steps into frame oddly. Testing digs into crashes, yet the goal is graceful reactions, never full stop.

Overall, checking the full setup shows the Face Recognition Attendance System works well as one complete unit. When this step passes, it means the system can move forward. Real world use becomes possible only after everything runs smoothly together.

8.2 TEST CASE

TestCase ID	Test Description	Input	Expected Output	Actual Output	Status
TC01	Verify camera initialization	Start system	Camera opens successfully	As expected	Pass
TC02	Capture live video feed	Live camera input	Continuous video frames	As expected	Pass
TC04	Detect face in normal lighting	Face image	Face detected correctly	As expected	Pass
TC05	Detect multiple faces	Group image	All faces detected	As expected	Pass
TC06	Detect face in low lighting	Face image	Face detected with acceptable accuracy	As expected	Pass
TC07	Recognize registered user	Known face	Correct identity predicted	As expected	Pass
TC08	Detect unregistered user	Unknown face	Marked as unknown	As expected	Pass
TC09	Recognition with facial expression change	Smiling/angled face	Correct recognition	As expected	Pass
TC10	Mark attendance for valid user	Recognized face	Attendance marked	As expected	Pass
TC11	Prevent duplicate attendance	Same face twice	Attendance marked once	As expected	Pass
TC12	Store date and time	Recognized face	Date & time recorded	As expected	Pass
TC13	Save attendance to CSV/database	Attendance data	Data saved correctly	As expected	Pass
TC14	Retrieve attendance records	Query records	Records displayed	As expected	Pass

TC16	End-to-end system execution	Live face input	Attendance marked successfully	As expected	Pass
TC17	System response time	Live input	Real-time processing	As expected	Pass
TC18	Error handling	Invalid input	No system crash	As expected	Pass

CHAPTER 9

CONCLUSION

The Face Recognise Attendance System that has been constructed in this project is an example of application of computer vision of high deep learning technologies that can effectively solve real-world administrative problems. Gone are the days of messy sign-in sheets. With faces as keys, showing up gets logged without touch or hassle. Picture pixels turning into proof - cameras spot who's there by studying shapes and shadows. Instead of fingerprints or swipe cards, eyes and bone structure do the work quietly. Learning happens through layers that mimic how brains see patterns. Mistakes fade when machines recognize people like humans do - but faster. What once needed paper trails now flows silently through smart software. Security tightens because faking a face grows harder than forging a signature. Time stamps lock in automatically, no buttons pressed. This shift didn't come overnight - it built on math, memory, and millions of images. Accuracy climbs when systems learn from more examples. Each check-in sharpens the model further. No extra gear. Just light, lenses, and logic.

One big reason the project worked well was using CNN to boost how accurately faces are recognized. Instead of needing handpicked traits, CNN finds key face details on its own from raw input - unlike older methods. Because of this, shifting light, varied expressions, or tilted heads cause less trouble than before. Thanks to these adjustments, running fast becomes possible, making live tracking realistic in schools, workplaces, and similar spots. By switching to automatic check-ins through face scans, fake entries drop sharply since copying someone else fails more often now. Fewer people need to get involved because the system runs on its own, cutting down busywork while getting more done. Starting tasks without physical contact plays well into clean conditions and keeps everyone safer, something any workplace today can't do without.

Every piece fits. Not just parts sitting together but working - data gathering flows into face spotting, which passes results to neural network checks, then neatly tucks outcomes into storage. Built separate yet linked, each block supports the next without mess. Records line up clean, stored so they can be found fast when needed later. Files shaped right mean reports form smoothly, analysis runs without hitches. Nothing extra, nothing missing - just pieces doing their job where they should.

CHAPTER 10

FUTURE ENHANCEMENTS

• There are AI-powered attendance programs which are beginning to make attendance easier. Alongside our system, a cloud platform for Student Affairs Management was established that covered the same system processes as Krisac University on KR-SIS.

• You can try to develop more elaborate depth learning models to raise accuracy. This development might be based on stronger CNN structures and involve pre-trained face recognition models.

• The system can also do better under poor light conditions, half-covered faces, or changes in typical facial appearance.

• By upgrading the recognition model, the system will also be able to support more users without performance deteriorating.

• Another major improvement that the future holds is enhancing security against spoofing attacks, such as a use case of photographs or recorded videos.

• Techniques like blink detection, facial movement analysis and 3D sensing are developed to ensure only live human faces are recognized.

• Cloud integration will be an important direction for future versions of the system to go in.

• Storing attendance data on a cloud platform will provide centralized access, data backups, and real-time monitoring in different locations.

• Cloud-based storage will be particularly useful for educational institutions with many campuses or branches.

• Developing a mobile or web-based application can improve accessibility for administrators and authorized users.

• Through these applay, users can view attendance record, generate reports, and receive notifications easily.

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Appendix A

BIBILOGRAPHY

1.Han, J., Kamber, M., and Pei, J., *Data Mining: Concept and Techniques*, Morgan Kaufmann Publisher, 3rd Edition, pp. 1–740, 2012.

2. Bishop, C. M., *Pattern Recognition and Machine Learning*, Springer, pp. 1–738, 2006.

3.Aggarwal, C. C., *Data Mining: The Textbook*, Springer, pp. 1–734, 2015.

4. Russell, S., and Norvig, P., *Artificial Intelligence: A Modem Approach*, Pearson Education, 3rd Edition, pp. 1–1152, 2010.

5. Géron, A., *Hands-On Machine Learning with Scikit-Learn and TensorFlow*, O'Reilly Media, 2nd Edition, pp. 1–856, 2019.

6. Paul Viola and Michael Jones Rafid object detection use boosted classifiers, 2001

7. Matthew Turk and Alex Pentland Face recognition using eigenfaces, 1991

8. Peter N. Belhumeur, João P. Hespanha, and David J. Kriegmans Eigenface for face recognition, 1997

Appendix B

USER MANUAL

1. INTRODUCTION

Security grows stronger since only real people get recognized - no duplicates sneak through. Each scan updates records fast, leaving behind old ways of signing in.

2. SYSTEM REQUIREMENTS

• A computer a laptop with a working camera
• Stable power connection
• Operating system such as Windows or Linux/Unix
• Python installed on the model
• Required Python lib for face recognition and images processing
• Database -storage system for storing attendance records

3. USER ROLES

The system supports the following user roles:

• Administrator
Responsibefor managing users, registering faces, and viewing attendance records.

• Authorized User
Can operate the system to mark attendance and view the reports as permitted.

4.START THE APPLICATION PROGRAM.

To start the Face Recognition Attendance System:

• Turn on the system and ensure the camera is connected
• Open the application program
• Run the main execution file
• Wait until the camera window opens and starts.

Once the camera starts, the system is ready to recognises faces.

5. FACE REGISTRATION PROCESS

Face registration must be completed before attendance can be marked.

• Open the face registration module
• Enter user details such as name or ID
• Stand in front of camera
• Ensure proper lighting and clear visibility of the face
• The system captures multiple face images
• Images are stored in the dataset for training

After registration, the face data is saved for recognition.

6. TRAINING THE SYSTEM

To improve recognition accuracy, the system needs to be trained.

• Run the training module
• The system processes stored face images
• Face features are extracted using the CNN model
• The trained model is saved for recognition

Training needs to be performed whenever new users are added.

7. ATTENDANCE MARKING

To mark attendance:

• Start the attendance system
• The camera detects face in real time
• Faces found the machine link to stored in record through comparison.

• Attendance details such as name, date, and time are recorded

8. VIEWING ATTENDANCE RECORDS

Attendance records can be viewed by authorized users.

• Open the attendance records section
• Select date or user if required
• Attendance details are displayed on the screen
• Records can be stored or exported if needed

9. ERROR HANDLING

If the system does not recognize a face:

• Ensure proper lighting
• Face should be clearly visible
• Avoid covering the face
• Ensure the user is registered
• Retrain the system if new data is added

10.SECURITY FEATURES

• Reduces proxy and unauthorized attendance
• Only registered users are recognized
• Data access is restricted to authorized users

11. System Shutdown

To safely close the system:

• Stop the attendance module
• Close the camera window
• Exit the application properly

12. Maintenance Guidelines

• Regular updateing the datasets
• Retrains the model when the new user are added
• Backup the attendance dat periodically
• Keep the systems environment clean and security

13. Conclusion

Starting off, the Face Recognition Attendnce System makes tracking who’s present straightforward. Because it works automatically, keeping logs correct takes almost no time at all. Following steps in this guide helps people get going without any confusion. Instead of guessing, workers rely on clear prompts to move forward. Each step fits into daily routines without slowing things down.